

FLIGHT SOFTWARE V1 — BALLOON LAUNCH SCHEDULE

From SIM to Sky: A Week-by-Week Plan, August 12 – November 1

Boards don't arrive until September — so every week before that is SIM-only, and hardware bring-up starts the instant boards land, not after software is "done."

Status: companion to `balloon_launch_plan.md` — that document defines what to build; this one maps it onto the calendar

Governing constraint: physical boards arrive in September (assumed ~Week 4 / Sep 8 — update the moment a real ship date is known)

Companion documents: `docs/balloon_launch_plan.md` , `docs/roadmap.md`

§ Assumptions Behind This Schedule

Update immediately if any of these turn out wrong — the whole schedule shifts with them.

- **Boards arrive by the end of Week 4 (~Sep 8).** If you get a firmer ship date, move the Week 4/5 boundary to match it — everything after that point is written relative to "hardware in hand," not a fixed date.
- **Phase B2 (subsystem scaffolding) is worked by multiple owners in parallel** once Phase B1 lands, per `balloon_launch_plan.md` 's parallelization map — this schedule doesn't fit in 12 weeks if one person does it serially.
- **Nov 1 is the flight-ready / go-no-go target**, not necessarily the literal launch day — `balloon_launch_plan.md` already recommends 2–3 weeks of buffer between "ready" and "launch." If the actual launch date is later than Nov 1, that gap is your buffer, not slack to spend on scope creep.

§ Progress Milestones

The clear markers, in order. Each one is a fact about the system, not a date on a calendar — use these to tell whether the schedule below is on track, not just whether the calendar pages have turned.

#	Milestone	Target week	Signal you've hit it
M0	Kickoff & hardware ordered	W1	Plan locked; every long-lead item (boards, GPS module, battery hardware) ordered <i>this week</i> , not "soon"
M1	Foundation complete	W3	balloon_launch_plan.md Phase B1 (roadmap.md 1.1–1.22) done; Foundation DoD dry run passes
M2	All-SIM full bus validated	W4	Every board — ADCS, EPS, Thermals, Camera, Comms — passes Phase 2 DoD in SIM
M3	Boards arrive / bring-up begins	W4 (assumption)	Hardware physically on hand; Phase B5 kicked off the same day, not the same week
M4	OBC Linux boots on real hardware	W6	PetaLinux/Yocto image boots on the real Zynq PS
M5	Comms real Zephyr build running	W6	STM32H743 boots Zephyr + libsatnogs-comms, runs its SIM-validated scaffolding logic for real
M6	Real I2C bus verified	W7	Real I2C traffic observed between two real boards
M7	First real cross-board CSP cycle	W7	A real command/telemetry round trip between OBC and one subsystem board, on hardware
M8	All boards individually bring-up complete	W9	Every subsystem board + Comms passes Phase 2 DoD on real hardware
M9	Real GPS integrated & field-tested	W9	Outdoor lock achieved, position visible in downlinked telemetry
M10	Real EPS battery telemetry integrated	W9	Voltage/current visible and plausible across a multi-hour bench run
M11	Full end-to-end system test passes	W10	All boards + real RF ground contact, simultaneously, on real hardware
M12	Flight readiness checklist complete	W11–12	Soak, range, cold, fault-recovery, power-budget, and onboard-logging checks all pass
M13	Go/No-Go: flight-ready	W12 (by Nov 1)	Final checklist signed off; dry run of the real flight-day commands executes clean

§ Week-by-Week Plan

PHASE: SIM-ONLY (NO HARDWARE YET) — WEEKS 1–4, AUG 12 – SEP 8

Week	Dates	Focus	Key tasks	Milestone(s)
------	-------	-------	-----------	--------------

W1	Aug 12–18	Kickoff; start Foundation	Order all long-lead hardware today . balloon_launch_plan.md 1.1–1.5 (directory/bus convention migration)	M0
W2	Aug 19–25	Foundation continues	1.6–1.10 (I2C SIM backend, ADCS+OBC ported and re-verified over it), 1.11–1.12 (shared envelope)	—
W3	Aug 26–Sep 1	Foundation finishes; scaffolding starts	1.13–1.22 (SPI peripheral pattern, scaffolding template, test harness, CI, Foundation DoD dry run). Start Phase B2 SIM work: 2.1–2.5 for EPS/Thermals/Camera, 2C.1–2C.2 for Comms	M1
W4	Sep 2–8	Full-SIM validation; hardware lands	Finish Phase B2 SIM work: 2.6–2.8 (EPS/Thermals/Camera), 2C.3–2C.6 (Comms), ADCS re-verification against the new I2C SIM backend (~3h). The moment boards arrive: kick off 6.1 (STM32 toolchain) and 6.2 (PetaLinux/Yocto — start this first, longest lead time in the plan)	M2, M3

PHASE: HARDWARE BRING-UP & INTEGRATION — WEEKS 5–9, SEP 9 – OCT 13

Week	Dates	Focus	Key tasks	Milestone(s)
W5	Sep 9–15	OBC + Comms bring-up begins	6.2 continues (PetaLinux/Yocto — expect this to spill past one session), 6.3 starts (Comms real Zephyr build). Start B3.1–B3.2 (GPS module confirmed, SPI driver) if OBC hardware allows	—
W6	Sep 16–22	OBC and Comms both boot for real	Finish 6.2 and 6.3. Start 6.6 (real I2C driver, both sides)	M4, M5
W7	Sep 23–29	First real bus traffic	Finish 6.6. First real I2C test between two boards. Begin 6.7 (ADCS bench bring-up — start with ADCS since its software is most mature)	M6, M7
W8	Sep 30–Oct 6	Per-board bring-up fans out	6.8 for EPS, Thermals, Camera (parallel across owners if the team allows). Continue B3.2–B3.3 (GPS wired into OBC telemetry)	—
W9	Oct 7–13	Bring-up wraps; balloon-critical sensors go live	Finish 6.8 (all boards). B3.4 (GPS field test). B3.5–B3.6 (real EPS battery telemetry). Phase B4 (CI + TO) should be substantially done by now — mostly software work that only needed real Comms to point at	M8, M9, M10

PHASE: FULL INTEGRATION & FLIGHT READINESS — WEEKS 10–12, OCT 14 – NOV 1

Week	Dates	Focus	Key tasks	Milestone(s)
W10	Oct 14–20	Everything, together, for real	6.9: full end-to-end system test — all boards + real RF ground contact, simultaneously. Budget real slack this week; this is the first time everything runs together and will surface integration bugs	M11
W11	Oct 21–27	Flight readiness checks	B6.1 (long-duration soak — start early in the week, it needs to run unattended), B6.2 (range/link test), B6.3 (cold-environment check), B6.5 (fault-recovery check), B6.6 (onboard logging backup)	—

W12	Oct 28–Nov 1	Final checks & go/no-go	B6.4 (power budget, can pull earlier if B3.6 finished sooner), any B6 items still open, B6.7 (final go/no-go checklist + dry run)	M12, M13
------------	--------------	-------------------------	---	-----------------

§ If Hardware Slips Past Early September

The single biggest risk to this schedule is boards arriving later than Week 4. If that happens, **don't try to compress Phase B5 — cut scope instead**, in this order (matching balloon_launch_plan.md's own must-have/should-have split):

1. **First to defer:** Camera and Thermals real bring-up. Both are scaffolding-only for this flight already (no real control logic) — a late arrival can leave them SIM-validated and physically un-flown if it comes to that, without breaking the flight's core purpose.
2. **Second to defer:** the Should-have items (real IMU/magnetometer logging on ADCS, real thermistor logging on Thermals) — cut these first, they were never must-have.
3. **Never cut, compress the schedule around instead:** OBC, Comms (with real RF), ADCS (as the bus-validation reference), EPS (with real battery telemetry), and GPS. These five are the actual must-have list from balloon_launch_plan.md — if the date gets tight, everything else bends around protecting these.
4. **If it's genuinely not going to fit:** raise that explicitly rather than quietly dropping Phase B6's flight-readiness checks to make room. A flight that skips the soak test or the fault-recovery check to hit a date is a worse outcome than a flight that slips a week or two.

Flight Software V1 — Balloon Launch Schedule. Companion to balloon_launch_plan.md; both are living documents — update the week boundaries the moment a real hardware ship date is known.